Introduction to Euclid's Geometry

The big idea

Around 300 BCE, the Greek mathematician **Euclid** organised all known geometry in his book *Elements*. His method: start from a few **definitions** and obvious truths (**axioms** and **postulates**), then **prove** everything else (**theorems**) logically. This is still how mathematics works.

Definitions

Euclid described basic ideas: "A point is that which has no part", "A line is breadthless length", "The ends of a line are points." Terms like point, line and plane are taken as **undefined**: we understand them without a formal definition.

Axioms (common notions)

These are assumptions used throughout mathematics, not just geometry:

1. Things equal to the same thing are equal to one another.
2. If equals are added to equals, the wholes are equal.
3. If equals are subtracted from equals, the remainders are equal.
4. Things that coincide with one another are equal.
5. **The whole is greater than the part.**
6. Things that are double of the same things are equal.
7. Things that are halves of the same things are equal.

Euclid's five postulates

These are assumptions specific to geometry:

1. A straight line may be drawn from any one point to any other point.
2. A terminated line can be produced indefinitely.
3. A circle can be drawn with any centre and any radius.
4. All right angles are equal to one another.
5. If a straight line falling on two straight lines makes the interior angles on the same side together less than two right angles, then the two lines, if produced indefinitely, meet on that side.

THE FIFTH POSTULATE

Its simpler equivalent (Playfair's axiom): **for every line l and every point P not on l, there is exactly one line through P parallel to l**. Equivalently: two distinct intersecting lines cannot both be parallel to the same line.

A THEOREM FROM THE AXIOMS

Two distinct lines cannot have more than one point in common. If they shared two points P and Q, there would be two different lines through P and Q, contradicting the axiom that a unique line passes through two distinct points.

AXIOM VS THEOREM

Axioms/postulates are accepted without proof. **Theorems** are statements that are **proved** using axioms, definitions and earlier theorems.

Quick recap

- Euclid's method: definitions + axioms + postulates ⇒ theorems by proof.
- Know the 7 axioms and 5 postulates (especially the 5th).
- Two distinct lines meet in at most one point.

Practice questions

1. Euclid's book of geometry is called
   (a) Principia (b) Elements (c) Geometria (d) Sulbasutras
2. "The whole is greater than the part" is one of Euclid's
   (a) axioms (b) postulates (c) theorems (d) definitions
3. According to the fifth postulate (Playfair's form), through a point not on a line we can draw
   (a) no parallel line (b) exactly one parallel line (c) two parallel lines (d) infinitely many parallel lines
4. Statements that are proved are called
   (a) axioms (b) postulates (c) theorems (d) definitions
5. Two distinct lines can have at most how many points in common?
   (a) 0 (b) 1 (c) 2 (d) infinitely many

Answers: 1. b 2. a 3. b 4. c 5. b